

LLM Benchmark

June 29, 2026

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Introduction

Goal and Keypoints

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Goal: build a reproducible benchmark for evaluating how LLMs solve PDDL planning problems and evaluate the reasoning capabilities.

Presentation structure

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Repository Structure

General Structure

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Repository Structure

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The repository is split into three main areas:

- **analysis:** notebooks, reports and generated figures.
- **archive:** reference papers, planning domains and PDDL support material.
- **Benchmark_Framework:** execution pipeline, tasks, model backends, validation and outputs.

LLM Benchmark/

```
├── analysis/
│   ├── domain_complexity/
│   ├── notebooks/
│   └── reports/
├── archive/
│   ├── Papers/
│   ├── Planning_Domains/
│   ├── References/
│   └── Quick_PDDL_Guide/
├── Benchmark_Framework/
│   └── ...
├── results/
└── README.md
```

General Structure

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`Benchmark_Framework` is the runnable benchmark layer. Only the core folders are shown here.

- *tasks*: selected PDDL domains and instances.
- *prompts* and *protocols*: prompt text, direct planning and repair settings.
- *models*: backend registries and adapter configuration.
- *runner*: suite and single-case execution logic.
- *evaluators*: parser, VAL validation, metrics and error taxonomy.
- *outputs*: raw, parsed and scored benchmark artifacts.
- *reporting*: post-run plots and JSON reports.
- *utils*: bundled VAL validator binaries.

`Benchmark_Framework/`

```
| evaluators/  
| models/  
| outputs/  
| prompts/  
| protocols/  
| reporting/  
| runner/  
| tasks/  
| utils/  
| run_benchmark.py
```

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Benchmark Framework Overview

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The framework is organized as a reproducible pipeline: the same task set can be tested with different prompts, protocols and model backends.

`tasks + prompts + protocols + model registries`



`run_benchmark.py`



`runner/run_suite.py → runner/run_case.py`



`model adapter → parser → VAL validator → metrics`



`outputs/raw + outputs/parsed + outputs/scored`

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Why This Structure

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- **Comparable runs:** every model is evaluated on the same task, protocol and validation matrix.
- **Backend isolation:** model adapters keep HuggingFace, NVIDIA API, Ollama and llama.cpp behind the same generation interface.
- **External validation:** VAL checks the generated plans independently from the model text.
- **Traceable artifacts:** raw generations, parsed plans and scored validation results are stored separately.
- **Post-run analysis:** reports and plots are generated from saved JSON artifacts, not from manual inspection.

Problem Classification

Classification Purpose

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Purpose: select a small but meaningful set of PDDL tasks from the available domains.

The classification has two complementary goals:

- assign a numerical difficulty score to each domain and instance;
- label each domain by its dominant reasoning type.

This makes the benchmark easier to interpret: model failures can be compared by both difficulty and planning style.

Implementation: Raw and Numerical Score

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Difficulty scores are computed with
`Benchmark_Framework/scripts/score_domains_complexity.py`.

The score combines features that increase planning complexity:

- number of objects
- possible grounded actions
- number of goals
- numeric goal conditions
- numeric functions
- numeric preconditions
- numeric effects
- metric optimization
- topology structure

Each instance receives a raw score, then scores are normalized within the domain and summarized at domain level.

Implementation: Raw and Numerical Score II

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The raw score is a weighted logarithmic combination, so no single count dominates the result:

$$RawScore = 0.45 \cdot \log(1 + Nactions) + 0.20 \cdot \log(1 + Ngoals) + 0.20 \cdot \log(1 + NumScore) + 0.15 \cdot Topology$$

$$NumScore = 1.0 \cdot NumEffects + 2.0 \cdot NumPreconditions + 3.0 \cdot NumGoalConditions + 1.0 \cdot HasMetric$$

The example shows the *counters* domain summary: instance scores plus aggregate raw-score statistics.

```
{  
  "domain": "counters",  
  "instances": 20,  
  "total_raw_score": 59.765337,  
  "min_raw_score": 1.820528,  
  "median_raw_score": 3.177554,  
  "mean_raw_score": 2.988267,  
  "max_raw_score": 3.679334,  
  "domain_difficulty_0_100": 18.130829  
},
```

Implementation: Reasoning Type

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Difficulty alone is not enough: two domains can be equally hard while requiring different reasoning skills.

We therefore assign each domain a dominant reasoning category:

- numerical optimization
- symbolic logic
- temporal planning
- logistics
- spatial planning

Mixed domains are labeled by the reasoning type that is most central to solving the problem.

Problems Chosen

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We selected six domains from the three reasoning categories with enough coverage: numerical optimization, logistics and spatial planning.

Hard Set

- settlersnumeric (numerical optimization)
- rover (logistics)
- block-grouping (spatial)

Easy Set

- fo-counters (numerical optimization)
- expedition (logistics)
- fo-sailing (spatial)

Temporal planning and symbolic logic were excluded from the final set because only one domain was available for each category.

Tools Used

Tools Used

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The benchmark relies on a small set of tools, each with a specific role:

- **Leonardo HPC by CINECA:** main compute platform for local model execution.
- **Hugging Face:** source for local inference setup and download of open-source models.
- **whichLLM:** model-selection support based on available hardware on Leonardo.
- **VAL validator:** external PDDL plan validator.
- **NVIDIA API:** alternative backend to use open-source models without a local powerful machine.

Both Leonardo and the NVIDIA API were used: Leonardo for controlled local execution with direct GPU access, and NVIDIA API as a hosted backend for additional model runs.

Selected Models

Selected Models

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Model selection was guided by two constraints:

- compatibility with the available Leonardo hardware;
- coverage of recent open LLM families available through Hugging Face.

The local evaluation shown in the results includes:

- Gemma 4 31B IT
- Nemotron 3 Nano 30B A3B
- Phi-4
- Qwen 3 6/27B

Recommended Models					
#	Model	Quant	Fit / VRAM	Speed	Published
1	deepseek-ai/DeepSeek-V4-Flash 280.0B (13.0Ba)	Q6_K	Full GPU 217.6 GB	13.5 tok/s ~	2026-04-22
2	XiaoMiMo/MiMo-V2.5 210.0B (19.0Ba)	Q6_K	Full GPU 238.1 GB	12.4 tok/s ~	2026-04-27
3	Qwen/Qwen3.6-27B 27.8B	Q6_K	Full GPU 24.4 GB	34.6 tok/s ~	2026-04-21
4	Qwen/Qwen3-235B-A22B 235.1B (22.0Ba)	Q6_K	Full GPU 181.7 GB	16.3 tok/s ~	2025-04-27
5	google/gemma-4-31B-it 32.7B	Q6_K	Full GPU 28.6 GB	29.4 tok/s ~	2026-03-11
6	Qwen/Qwen3-Next-80B-A3B-Instruct 81.3B (3.0Ba)	Q8_0	Full GPU 81.8 GB	32.5 tok/s ~	2025-09-09
7	openai/gpt-oss-120b 120.4B (5.16a)	Q6_K	Full GPU 92.8 GB	31.9 tok/s ~	2025-08-04
8	MiniMaxAI/MiniMax-M2.5 226.7B (10.8Ba)	Q6_K	Full GPU 175.3 GB	16.8 tok/s ~	2026-02-12
9	Qwen/Qwen3-30B-A3B 30.5B (3.0Ba)	Q8_0	Full GPU 31.6 GB	86.6 tok/s ~	2025-04-27
10	mistralai/Mistral-Large-Instruct-2411 122.6B	Q5_K_M	Full GPU 98.3 GB	9.6 tok/s ~	2024-11-14

Outputs and Evaluation

Output Structure

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Each benchmark case produces three aligned JSON artifacts:

- **Raw:** prompt messages, model output and provider-side reasoning text when available.
- **Parsed:** extracted PDDL actions, final plan text and formatting issues.
- **Scored:** VAL validation result, solved flag, iterations used and repair feedback.

The suite also writes `suite_result.json`, which summarizes the run matrix and is the entry point for post-run analysis.

This split keeps generation, extraction and evaluation independent, so parser errors and planning failures can be diagnosed separately.

Evaluation Metrics Scope

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The metrics are designed to answer three different questions:

- **Did the model solve the planning task?**
- **Was the produced action sequence grounded in the PDDL domain?**
- **Did retries improve the answer or only resample attempts?**

For this reason, the evaluation does not rely only on final success. It combines success, executability, hallucination, precondition awareness and retry behavior.

Core Metrics

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- **SR**: fraction of instances solved by the model.
- **FASR**: first-attempt success rate; measures zero-shot planning ability.
- **IWSR**: iteration-weighted success rate; rewards valid plans found with fewer retries.
- **Retry Gap**: $SR - FASR$; high values mean the model depends on repair attempts.
- **Executability Ratio**: fraction of the plan executable before the first precondition failure.
- **Hallucination Rate**: fraction of actions or objects not grounded in the PDDL files.
- **PAS**: precondition awareness score; separates sequencing errors from fabricated states.
- **CoTaL**: alignment between reasoning actions and the final extracted plan when reasoning is available.

Cross-Domain and Cross-Model Analysis

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- **Within-domain ranking:** compares models inside each domain using normalized metric values.
- **Cross-domain consistency:** checks whether a model is robust across different reasoning categories.
- **Failure mode taxonomy:** combines hallucination, executability and PAS to identify why plans fail.
- **Difficulty analysis:** compares easy, medium and hard instances to see where performance degrades.

Composite Planning Score

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The Composite Planning Score summarizes complementary planning signals in one value:

$$PS = 0.25 \cdot FASR + 0.20 \cdot IWSR + 0.20 \cdot ER + 0.20 \cdot (1 - Hallucination) + 0.15 \cdot PAS$$

- **FASR** has the highest weight because it cannot be inflated by retries.
- **IWSR** and **ER** capture repair efficiency and structural plan quality.
- **Grounding** is represented by inverse hallucination rate.
- **PAS** adds information about local precondition reasoning.

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Evaluation Snapshot

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The plots shown here come from the generated report `results/LLM_classification_test`.

- **Rows:** 267 evaluated task instances.
- **Models:** Gemma, Nemotron, Phi-4 and Qwen.
- **Domains:** block-grouping, expedition, fo-counters, fo-sailing, rover and settlersnumeric.
- **Protocol:** iterative repair.
- **Difficulty tiers:** easy, medium and hard.

Success and Hallucination

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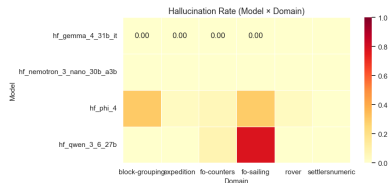
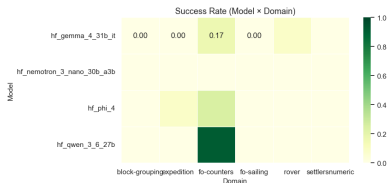
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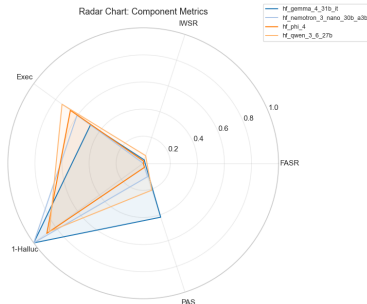
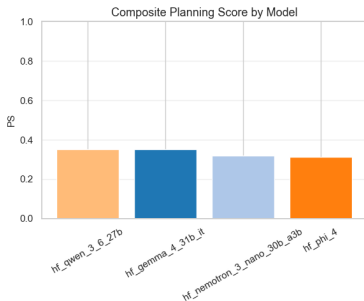


Final success and grounding errors must be read together: a model can use legal-looking actions while still failing the global goal.

Composite Score and Capability Profile

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The composite score gives a compact ranking; the radar chart shows which capabilities drive each score.

Conclusions

Benchmark Recap

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- The project provides a reproducible benchmark for PDDL planning tasks.
- The repository separates task selection, benchmark execution, validation and post-run analysis.
- The framework stores raw generations, parsed plans and scored validation results as separate artifacts.
- VAL is used as an external validator, so final scoring does not depend only on model wording or parser interpretation.

LLM Reasoning Capabilities

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- The tested LLMs show partial planning ability, but robust goal-directed planning remains difficult.
- Models often produce action sequences that are locally plausible or partially executable, while still failing to reach the global goal.
- First-attempt success is low, so iterative repair is useful but does not fully solve the planning gap.
- Failures are not all the same: the metrics separate grounding errors, precondition mistakes, retry dependence and reasoning-plan alignment.
- Chain-of-thought can be useful diagnostically, but it must be checked against the final extracted plan and VAL validation.

Future Work

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- Extend the benchmark with more temporal planning and symbolic logic domains.
- Run a larger model matrix using both Leonardo and NVIDIA API backends.
- Improve prompt and repair-feedback strategies to test whether models can correct planning failures rather than resample.
- Deepen the analysis of reasoning traces with CoTaL and failure-mode taxonomy.
- Add more result summaries that compare reasoning profiles across domains, difficulty levels and model families.